

CONSTANTS	$F_s = 32,000$ psi	allowable steel stress, Grade 60 (§8.3.3.1)
	$F_b = 0.45f'_m$	allowable flexural compression, reinforced (§8.3.4.2.2)
	$E_m = 900f'_m$	concrete masonry modulus; clay: $700f'_m$ (§4.2.2)
	$E_s = 29,000,000$ psi	steel modulus (§4.2.2)

1 SHEAR (ASD)

WHERE: TMS 402-16 §8.3.5 · ANSWERS: STEEL-ONLY / MASONRY-ONLY / COMBINED SHEAR CAPACITY · APPLIES: BEAMS, WALLS IN-PLANE, PIERS

$$f_v = \frac{V}{A_{nv}} \leq F_v = (F_{vm} + F_{vs}) \gamma_g \quad (1)$$

f_v	unit shear stress
A_{nv}	net shear area
F_v	total allowable unit shear stress
F_{vm}	masonry share
F_{vs}	shear-reinforcement share
γ_g	0.75 partially grouted shear wall, else 1.0

$$F_{vm} = \frac{1}{2} \left[4.0 - 1.75 \frac{M}{Vd_v} \right] \sqrt{f'_m} + 0.25 \frac{P}{A_{nv}} \quad (2)$$

d_v	actual member depth in direction of shear
$M/(Vd_v)$	taken ≤ 1.0 ; axial term P zero for beams

$$F_{vs} = 0.5 \frac{A_v F_s d_v}{A_{nv} s} \quad (3)$$

A_v shear reinforcement area per spacing s

LIMITS

$M/(Vd_v) \leq 0.25$	$F_v \leq 3\sqrt{f'_m} \gamma_g$
$M/(Vd_v) \geq 1.0$	$F_v \leq 2\sqrt{f'_m} \gamma_g$
between	interpolate
special reinforced shear wall	F_{vm} coefficient $\frac{1}{2} \rightarrow \frac{1}{4}$; design for $1.5\times$ seismic V

2 FLEXURE (ASD)

WHERE: TMS 402-16 §8.3.4 · ANSWERS: BEAM / WALL MOMENT CAPACITY · APPLIES: BEAMS, LINTELS, WALLS OUT-OF-PLANE

$$n = \frac{E_s}{E_m} \quad \rho = \frac{A_s}{bd} \quad k = \sqrt{(n\rho)^2 + 2n\rho} - n\rho \quad j = 1 - \frac{k}{3} \quad (4)$$

n	modular ratio
k	neutral-axis depth ratio
j	moment-arm ratio

$$M \leq \min \left(A_s F_s j d, \frac{1}{2} F_b k j b d^2 \right) \quad (5)$$

$A_s F_s j d$	steel governs
$\frac{1}{2} F_b k j b d^2$	masonry governs

3 AXIAL + INTERACTION (ASD)

WHERE: TMS 402-16 §8.2.4 (UNREINFORCED) · ANSWERS: UNREINFORCED WALL AXIAL CAPACITY, P+M UNITY CHECK

$$F_a = \frac{1}{4} f'_m \left[1 - \left(\frac{h}{140r} \right)^2 \right] \quad \left(\frac{h}{r} \leq 99 \right) \quad F_a = \frac{1}{4} f'_m \left(\frac{70r}{h} \right)^2 \quad \left(\frac{h}{r} > 99 \right) \quad (6)$$

F_a	allowable axial compressive stress
h	effective height
r	radius of gyration

$$\frac{f_a}{F_a} + \frac{f_b}{F_b} \leq 1 \quad P \leq \frac{1}{4}P_e = \frac{1}{4} \frac{\pi^2 E_m I_n}{h^2} \left(1 - 0.577 \frac{e}{r}\right)^3 \quad (7)$$

f_a, f_b computed axial, flexural stresses
 F_b here $\frac{1}{3}f'_m$ unreinforced (§8.2.4.2), not $0.45f'_m$
 P_e Euler buckling load; unreinforced check only
 I_n net moment of inertia
 e eccentricity

4 SHEAR + FLEXURE (STRENGTH DESIGN)

WHERE: TMS 402-16 §9.3 · ANSWERS: ϕV_n SHEAR / ϕM_n MOMENT (SD) · ϕ : SHEAR 0.80, FLEXURE 0.90 (§9.1.4)

$$V_u \leq \phi V_n = \phi (V_{nm} + V_{ns}) \gamma_g \quad (8)$$

V_{nm}, V_{ns} nominal masonry, reinforcement shares

$$V_{nm} = \left[4.0 - 1.75 \frac{M_u}{V_u d_v}\right] A_{nv} \sqrt{f'_m} + 0.25 P_u \quad V_{ns} = 0.5 \frac{A_v}{s} f_y d_v \quad (9)$$

$M_u/(V_u d_v)$ taken ≤ 1.0

$$a = \frac{A_s f_y + P_u}{0.80 f'_m b} \quad M_n = (A_s f_y + P_u) \left(d - \frac{a}{2}\right) \quad (10)$$

a equivalent stress-block depth ($0.80f'_m$, §9.3.2)
 ϵ_{mu} 0.0025 CMU, 0.0035 clay (§9.3.2)

LIMITS

$M_u/(V_u d_v) \leq 0.25$ $V_n \leq 6A_{nv} \sqrt{f'_m} \gamma_g$
 $M_u/(V_u d_v) \geq 1.0$ $V_n \leq 4A_{nv} \sqrt{f'_m} \gamma_g$
 between interpolate
 special wall capacity design shear $\geq$ shear at $1.25M_n$, need not exceed $2.5V_u$ (§7.3.2.6.1)

5 ANCHOR BOLTS (ASD)

WHERE: TMS 402-16 §8.1.3 · ANSWERS: HEADED-BOLT TENSION / SHEAR / COMBINED CAPACITY · MINIMUM MODE GOVERNS · BENT-BAR: ADD PULLOUT (EQ 8-3)

$$B_a = \min \left(B_{ab} = 1.25 A_{pt} \sqrt{f'_m}, \quad B_{as} = 0.6 A_b f_y \right) \quad (11)$$

B_a allowable tension
 B_{ab} masonry breakout
 B_{as} steel yield
 A_{pt} projected tension breakout area
 A_b bolt cross-sectional area

$$B_v = \min \left(B_{vb} = 1.25 A_{pv} \sqrt{f'_m}, \quad B_{vc} = 350 \sqrt[4]{f'_m A_b}, \quad B_{vpry} = 2.0 B_{ab}, \quad B_{vs} = 0.36 A_b f_y \right) \quad (12)$$

B_v allowable shear
 B_{vb} masonry breakout
 B_{vc} masonry crushing
 B_{vpry} pryout
 B_{vs} steel yield
 A_{pv} projected shear breakout area

$$\frac{b_a}{B_a} + \frac{b_v}{B_v} \leq 1 \quad (13)$$

b_a, b_v applied tension, shear

6 WORKED EXAMPLES

EX 1 · ASD SHEAR, MASONRY ONLY (§8.3.5.1.2) · f'_m 2000, A_{nv} 732 IN², P 40 K, $M/(Vd_v)$ 1.0

$$F_{vm} = \frac{1}{2} [4.0 - 1.75(1.0)] \sqrt{2000} + 0.25 \frac{40,000}{732} = 50.3 + 13.7 = 64.0 \text{ psi}$$

$$F_v \leq 2\sqrt{f'_m} \gamma_g = 89.4 \text{ psi } (\gamma_g = 1.0) \text{ — OK} \quad V = 64.0(732) = 46.8 \approx 47 \text{ kips}$$

TRAPS: drop axial term $\rightarrow 37 \cdot$ axial inside the $\frac{1}{2}$ bracket $\rightarrow 42$

EX 2 · ASD FLEXURE (§8.3.4) · f'_m 1500, f_y 60 KSI, b 7.625, d 20, A_s 0.88 IN²

$$E_m = 900f'_m; \quad n = \frac{29,000}{1350} = 21.5; \quad \rho = \frac{0.88}{7.625(20)} = 0.00577; \quad n\rho = 0.124$$

$$k = \sqrt{(n\rho)^2 + 2n\rho} - n\rho = 0.389; \quad j = 1 - k/3 = 0.870$$

$$M_s = A_s F_s j d = 40.8 \text{ k-ft}; \quad M_m = \frac{1}{2} F_b k j b d^2 = 29.1 \text{ k-ft} \Rightarrow \text{masonry governs, 29}$$

TRAPS: report steel value $\rightarrow 41 \cdot$ average of the two $\rightarrow 35$

EX 3 · UNREINFORCED UNITY (§8.2.4) · f'_m 1500, h 12 FT, r 2.2, f_a 130, f_b 200 PSI

$$h/r = 65.5 \leq 99 \Rightarrow F_a = \frac{1}{4} f'_m \left[1 - \left(\frac{144}{308} \right)^2 \right] = 293 \text{ psi (Eq 8-11)}$$

$$\frac{f_a}{F_a} + \frac{f_b}{F_b} = \frac{130}{293} + \frac{200}{500} = 0.44 + 0.40 = 0.84 \quad F_b = \frac{1}{3} f'_m = 500 \text{ psi, NOT } 0.45 f'_m$$

TRAPS: reinforced F_b 675 $\rightarrow 0.74 \cdot$ wrong slenderness branch $\rightarrow 0.70 \cdot$ axial alone $\rightarrow 0.44$

EX 4 · SD SHEAR, SPECIAL WALL (§9.3.4.1.2) · f'_m 3000, A_{nv} 549 IN², P_u 10 K, #4 @ 16 IN

$$M_u/(V_u d_v) = 16/6 = 2.67 \rightarrow \text{take 1.0}$$

$$V_{nm} = 2.25(549)\sqrt{3000} + 0.25(10,000) = 70.2 \text{ k}; \quad V_{ns} = 0.5 \frac{0.20}{16} (60,000)(72) = 27.0 \text{ k}$$

$$V_n = 97.2 \text{ k} \leq 4A_{nv}\sqrt{f'_m} = 120.3 \text{ k — OK}; \quad \phi V_n = 0.80(97.2) = 77.7 \approx 78 \text{ kips}$$

TRAPS: ϕV_{nm} only $\rightarrow 56 \cdot V_{nm}$ alone $\rightarrow 70 \cdot$ forgot $\phi \rightarrow 97$

EX 5 · SD FLEXURE (§9.3.2) · f'_m 2000, f_y 60 KSI, b 7.625, d 20, A_s 0.88 IN²

$$a = \frac{A_s f_y}{0.80 f'_m b} = \frac{52,800}{12,200} = 4.33 \text{ in.} \quad M_n = A_s f_y \left(d - \frac{a}{2} \right) = 78.5 \text{ k-ft}$$

$$\phi M_n = 0.90(78.5) = 70.6 \approx 71 \text{ k-ft} \quad (\phi = 0.90 \text{ flexure, §9.1.4.1})$$

TRAPS: forgot $\phi \rightarrow 79 \cdot$ lever arm $d - a \rightarrow 62 \cdot A_s f_y d$ alone $\rightarrow 88$